

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:50

Annexure A

Scenario-Based

Code of Conduct:

*I, K Suthesika (192511070) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

K Suthesika

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context(2)	Shows good understanding with minor gaps(1.5)	Partial understanding; misses key aspects(1)	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved(2)	Identifies most key issues with minor omissions(1.5)	Identifies limited issues; some irrelevant points(1)	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario(2.5)	Applies concepts with minor errors(2)	Limited or partially correct application(1.5)	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale(2)	Makes reasonable decisions with some justification(1.5)	Makes weak decisions with limited justification(1)	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented(1.5)	Generally clear with minor issues(1)	Somewhat unclear or poorly structured(0.5)	Disorganized and difficult to understand

1.A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

- 1. Determine the network address and broadcast address for each department.**
- 2. Identify the valid host IP address range for each subnet.**
- 3. Calculate the number of usable host addresses available in each subnet.**
- 4. Determine the remaining unused IP address range within the given network block.**

Given

Department	Subnet	Total addresses	Usable hosts
School of Computing	/25	128	126
School of Electronics	/26	64	62
School of Mechanical Engineering	/27	32	30

1. School of Computing — /25

/25 means the subnet mask is: **255.255.255.128**

The /24 network is divided into two blocks:

- 192.168.100.0 – 192.168.100.127

- 192.168.100.128 – 192.168.100.255

So we assign the first block.

- **Network address:** 192.168.100.0
- **First usable host:** 192.168.100.1
- **Last usable host:** 192.168.100.126
- **Broadcast address:** 192.168.100.127
- **Usable hosts:** 126

2. School of Electronics — /26

/26 means: **255.255.255.192**

Starting after 192.168.100.127, the next available block is:

192.168.100.128 – 192.168.100.191

Therefore:

- **Network address:** 192.168.100.128
- **First usable host:** 192.168.100.129
- **Last usable host:** 192.168.100.190
- **Broadcast address:** 192.168.100.191
- **Usable hosts:** 62

3. School of Mechanical Engineering — /27

/27 means:

255.255.255.224

The next available block is:

192.168.100.192 – 192.168.100.223

Therefore:

- **Network address:** 192.168.100.192
- **First usable host:** 192.168.100.193
- **Last usable host:** 192.168.100.222
- **Broadcast address:** 192.168.100.223
- **Usable hosts:** 30

4. Remaining unused IP address range

The original /24 block contains:

192.168.100.0 – 192.168.100.255

The allocated ranges end at:

192.168.100.223

Therefore, the remaining unused range is:

192.168.100.224 – 192.168.100.255

This gives **32 total IP addresses** remaining.

Final Answer

Department	Network Address	Valid Host Range	Broadcast Address	Usable Hosts
Computing /25	192.168.100.0	192.168.100.1 – 192.168.100.126	192.168.100.127	126
Electronics /26	192.168.100.128	192.168.100.129 – 192.168.100.190	192.168.100.191	62
Mechanical /27	192.168.100.192	192.168.100.193 – 192.168.100.222	192.168.100.223	30

Remaining unused IP range: 192.168.100.224 – 192.168.100.255

Remaining addresses: 32

2.A multinational software company is expanding its headquarters and has been assigned the private IP block 10.10.0.0/16. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → /24

Quality Assurance → /25

Human Resources → /26

Executive Management → /27

As the network administrator, perform the following tasks:

- 1. Determine the network address and broadcast address for each department.**
- 2. Identify the valid range of host IP addresses.**
- 3. Calculate the number of usable hosts in each subnet.**
- 4. Identify the unused IP address range remaining within the allocated IP block.**

Given

- Network block: 10.10.0.0/16
- Total addresses in /16 = 65,536
- Address range = 10.10.0.0 – 10.10.255.255

1. Software Development — /24

Subnet mask: **255.255.255.0**

A /24 contains **256 addresses**.

Starting address: 10.10.0.0

- **Network address:** 10.10.0.0
- **First host:** 10.10.0.1
- **Last host:** 10.10.0.254
- **Broadcast:** 10.10.0.255
- **Usable hosts:** $256 - 2 = 254$

2. Quality Assurance — /25

Subnet mask: **255.255.255.128**

A /25 contains **128 addresses**.

Next available address after the previous subnet is 10.10.1.0.

- **Network address:** 10.10.1.0
- **First host:** 10.10.1.1
- **Last host:** 10.10.1.126
- **Broadcast:** 10.10.1.127
- **Usable hosts:** $128 - 2 = 126$

3. Human Resources — /26

Subnet mask: **255.255.255.192**

A /26 contains **64 addresses**.

Next available address: 10.10.1.128.

- **Network address:** 10.10.1.128
- **First host:** 10.10.1.129
- **Last host:** 10.10.1.190
- **Broadcast:** 10.10.1.191
- **Usable hosts:** $64 - 2 = 62$

4. Executive Management — /27

Subnet mask: **255.255.255.224**

A /27 contains **32 addresses**.

Next available address: 10.10.1.192.

- **Network address:** 10.10.1.192
- **First host:** 10.10.1.193
- **Last host:** 10.10.1.222
- **Broadcast:** 10.10.1.223
- **Usable hosts:** $32 - 2 = 30$

5. Unused IP address range

The original /16 block ends at:

10.10.255.255

The last allocated subnet ends at:

10.10.1.223

Therefore, the remaining unused range is:

10.10.1.224 – 10.10.255.255

Total unused addresses:

$65,536 - (256 + 128 + 64 + 32) = 65,056$ addresses

Final Answer

Department	Subnet	Network Address	Valid Host Range	Broadcast Address	Usable Hosts
Software Development	/24	10.10.0.0	10.10.0.1 – 10.10.0.254	10.10.0.255	254
Quality Assurance	/25	10.10.1.0	10.10.1.1 – 10.10.1.126	10.10.1.127	126
Human Resources	/26	10.10.1.128	10.10.1.129 – 10.10.1.190	10.10.1.191	62
Executive Management	/27	10.10.1.192	10.10.1.193 – 10.10.1.222	10.10.1.223	30

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

- 1. Convert the given IPv6 address into its compressed notation.**
- 2. Expand the compressed address back into its full hexadecimal representation.**
- 3. Identify the network prefix using the /64 prefix length.**
- 4. Calculate the total number of host addresses supported within the subnet.**
- 5. the total number of possible host addresses available within this network.**

Given IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

An IPv6 address has 8 groups, each containing 4 hexadecimal digits.

1. Compressed notation

Given:

2001:0db8:abcd:0000:0000:0000:0000:0000

We can remove leading zeros from each group:

2001:db8:abcd:0:0:0:0:0

The consecutive groups of zeros can be replaced by :: **once**.

Therefore:

Compressed IPv6 address = 2001:db8:abcd::/64

2. Expanded/full hexadecimal representation

The compressed address:

2001:db8:abcd::/64

is expanded by restoring the five omitted 0000 groups:

2001:0db8:abcd:0000:0000:0000:0000/64

So the full representation is:

2001:0db8:abcd:0000:0000:0000:0000

3. Network prefix using /64

/64 means the **first 64 bits** are the network portion.

Since each hexadecimal group contains 16 bits:

64 bits = 4 hexadecimal groups

Therefore, the network prefix is:

2001:0db8:abcd:0000::/64

Or commonly written as:

2001:db8:abcd::/64

The first four groups are the network prefix:

2001 : 0db8 : abcd : 0000

The remaining four groups are available for host/interface addresses.

4 & 5. Total number of possible host addresses

IPv6 has **128 bits** in total.

Prefix length = **64 bits**

Therefore, host/interface bits:

128 – 64 = 64 bits

Number of possible addresses:

2^{64}

= 18,446,744,073,709,551,616

So the subnet supports:

18,446,744,073,709,551,616 possible IPv6 addresses

Unlike IPv4, IPv6 does **not** reserve a network address and broadcast address in the same way.

Therefore, essentially the entire 2^{64} space is available for interface addresses.

Final Answer

Question	Answer
Given address	2001:0db8:abcd:0000:0000:0000:0000/64
1. Compressed notation	2001:db8:abcd::/64
2. Expanded notation	2001:0db8:abcd:0000:0000:0000:0000/64
3. Network prefix	2001:db8:abcd::/64
4. Host bits	64 bits
5. Total possible host addresses	$2^{64} = 18,446,744,073,709,551,616$

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

- 1. The destination subnet to which the packet belongs.**
- 2. The matching routing table entry.**
- 3. The next-hop network/interface used to forward the packet.**
- 4. The default route that should be used if the destination does not match any routing table entry.**

Given Routing Table

Routing Entry	Subnet
192.168.1.0/24	192.168.1.0 – 192.168.1.255
192.168.2.0/24	192.168.2.0 – 192.168.2.255
192.168.3.0/24	192.168.3.0 – 192.168.3.255

Destination IP: **192.168.2.45**

1. Destination subnet

The first three octets are: 192.168.2

Therefore, 192.168.2.45 belongs to: **192.168.2.0/24**

Valid host range is: 192.168.2.1 – 192.168.2.254

2. Matching routing table entry

The destination 192.168.2.45 matches: **192.168.2.0/24**

So the router selects: **192.168.2.0/24**

This is an exact **/24 match**.

3. Next-hop network/interface

The routing table given does **not specify an actual next-hop IP address or interface name**. Since 192.168.2.0/24 is a directly connected network, the router would normally forward the packet through the **interface connected to the 192.168.2.0/24 network**.

So:

Next-hop/interface → Interface connected to 192.168.2.0/24

The router will use ARP to determine the destination device's MAC address in IPv4 and then forward the frame.

4. Default route

If the destination does not match any of the listed routes, the router uses a **default route**.

The default route is:

0.0.0.0/0

It means "**any IPv4 destination**."

However, the actual next-hop address/interface for the default route is **not provided in the question**, so it cannot be determined from the given routing table.

Final Answer

Task	Answer
1. Destination subnet	192.168.2.0/24
2. Matching routing entry	192.168.2.0/24
3. Next-hop/interface	Interface connected to 192.168.2.0/24
4. Default route	0.0.0.0/0

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → 192.168.10.0/26

Finance LAN → 192.168.10.64/26

The router interfaces are configured with:

Administration Gateway → 192.168.10.1

Finance Gateway → 192.168.10.65

As the network administrator, perform the following tasks:

- 1. Determine the network address and broadcast address for each LAN.**
- 2. Identify the valid host IP address range.**
- 3. Suggest suitable IP addresses for three computers in each LAN.**
- 4. Identify the default gateway that hosts in each LAN should use for communication.**

1. Network and Broadcast Addresses

Administration LAN — 192.168.10.0/26

A /26 subnet has **64 total addresses**.

Range: 192.168.10.0 – 192.168.10.63

- **Network address:** 192.168.10.0
- **Broadcast address:** 192.168.10.63
- **Valid host range:** 192.168.10.1 – 192.168.10.62

The gateway 192.168.10.1 is within this valid host range.

Finance LAN — 192.168.10.64/26

The next /26 block starts at 192.168.10.64.

Range:

192.168.10.64 – 192.168.10.127

- **Network address:** 192.168.10.64
- **Broadcast address:** 192.168.10.127
- **Valid host range:** 192.168.10.65 – 192.168.10.126

The gateway 192.168.10.65 is within this valid host range.

2. Valid Host IP Ranges

LAN	Network Address	Valid Host Range	Broadcast
Administration	192.168.10.0	192.168.10.1 – 192.168.10.62	192.168.10.63
Finance	192.168.10.64	192.168.10.65 – 192.168.10.126	192.168.10.127

Each /26 subnet provides:

$64 - 2 = 62$ usable host addresses

3. IP addresses for three computers

Administration LAN

Gateway = 192.168.10.1

Suitable computer IPs:

- **Computer 1:** 192.168.10.2
- **Computer 2:** 192.168.10.3
- **Computer 3:** 192.168.10.4

Subnet mask for all:

255.255.255.192

Finance LAN

Gateway = 192.168.10.65

Suitable computer IPs:

- **Computer 1:** 192.168.10.66
- **Computer 2:** 192.168.10.67
- **Computer 3:** 192.168.10.68

Subnet mask:

255.255.255.192

4. Default Gateway

Hosts should use the router interface belonging to their own LAN as the default gateway.

- **Administration computers → 192.168.10.1**

- **Finance computers → 192.168.10.65**

The router then forwards traffic between the two different subnets.

Final Configuration

Device/LAN	IP Address	Subnet Mask	Default Gateway
Admin Gateway	192.168.10.1	255.255.255.192	—
Admin PC1	192.168.10.2	255.255.255.192	192.168.10.1
Admin PC2	192.168.10.3	255.255.255.192	192.168.10.1
Admin PC3	192.168.10.4	255.255.255.192	192.168.10.1
Finance Gateway	192.168.10.65	255.255.255.192	—
Finance PC1	192.168.10.66	255.255.255.192	192.168.10.65
Finance PC2	192.168.10.67	255.255.255.192	192.168.10.65
Finance PC3	192.168.10.68	255.255.255.192	192.168.10.65